

WACE Mathematics Specialist · Cheat Sheet (Patterns + Notes)

General rules: principal argument $-\pi < \theta \leq \pi$ · Write mathematical statements, not calculator language · When explaining, don't write "it"; state which quantity (e.g. "SD of the sample mean") · Add absolute value signs to $\ln|\dots|$ (can omit when the argument is always positive) · Indefinite integrals + c · Rates of change with units · Expand brackets and watch for sign errors when copying (repeatedly noted in yearly reports) · Memorise exact trigonometric values for special angles · When a question asks prove / show, use algebra; "CAS says no solution" is not a proof · State the conclusion clearly

1 Complex numbers

① Form conversion and operations

- Division: multiply the numerator and denominator by the conjugate of the denominator. Powers and roots: first write as $r \operatorname{cis} \theta$, then use De Moivre $((\sqrt{3} - i)^5 = 32 \operatorname{cis}(-5\pi/6) = -16\sqrt{3} - 16i)$.
- Δ Third quadrant: the principal argument of $-1 - i$ is $-3\pi/4$, not $\pi/4$, and it cannot be written as $5\pi/4$. Draw the diagram first to determine the quadrant.
- Given $z = 3 + bi$, $\tan \theta = 2 \Rightarrow b = 6$, $r = 3\sqrt{5}$.

② Solving $z^n = w$ ★ appears every year

- Write w as $R \operatorname{cis} \varphi$ (modulus 1 mark, argument 1 mark) $\rightarrow z = R^{1/n} \operatorname{cis}((\varphi + 2k\pi)/n)$, $k = 0, \dots, n - 1$ (general formula 1 mark) $\rightarrow$ list **all** n roots within the range required by the question, spaced $2\pi/n$ apart.
- Example: $z^4 = -16i \Rightarrow 2 \operatorname{cis}(-5\pi/8)$, $2 \operatorname{cis}(-\pi/8)$, $2 \operatorname{cis}(3\pi/8)$, $2 \operatorname{cis}(7\pi/8)$.
- Roots of $z^n = 1$: equally spaced on the unit circle (explanation 2 marks: radius 1, equal spacing); product of all roots = $(-1)^{n+1}$; $z^{43} = 1$ and $z^{47} = 1$ have only the common root 1 (43 and 47 are coprime).
- z^n is a positive real number: $n \cdot \arg z$ is an integer multiple of 2π ($z = 4 \operatorname{cis}(5\pi/6) \Rightarrow$ smallest $n = 12$).

③ Complex roots of polynomials

- "Show $(z - \alpha)$ is a factor": calculate $P(\alpha)$, **write out every term**, obtain 0 (key: not just the statement).
- Real coefficients $\Rightarrow \bar{\alpha}$ is also a root $\Rightarrow (z - \alpha)(z - \bar{\alpha}) = z^2 - 2\operatorname{Re}(\alpha)z + |\alpha|^2 \rightarrow$ use division or compare coefficients to find the remaining factor $\rightarrow$ solve the quadratic.
- Example: $P(2 + 4i) = 0 \Rightarrow$ factor $z^2 - 4z + 20$, remaining $z^2 - 2z + 3 \Rightarrow z = 1 \pm \sqrt{2}i$. $\Delta z^2 - 2z + 3$ cannot be factorised as $(z - 3)(z + 1)$ (2021 report).
- Given two complex roots, work backwards to find coefficients: the two conjugate pairs each give a quadratic factor, then use the constant term to find the real root.

④ Geometric meaning of multiplication

- Template (1 mark each): "Multiplication by $w = r \operatorname{cis} \theta$ dilates the modulus by a factor r and rotates anticlockwise by θ (clockwise by $|\theta|$ if $\theta < 0$) about the origin." Multiply by z^{-1} : take the reciprocal of the scale factor and reverse the direction of rotation.
- Find n such that zw^n lies in the second quadrant: rotate by $\arg w$ each time; if $|w| = 1$ there is a period, so give all answers, e.g. $n = 3 + 12k$, $4 + 12k$, $5 + 12k$.

⑤ Loci: graph from equation

- $|z - a| = r$: circle. $|z - a| = |z - b|$: perpendicular bisector

of a and b . $\arg(z - a) = \theta$: ray, with **open endpoint at a** .

- $|z - a| + |z - b| = k$: if k is greater than $|a - b|$ it is an ellipse; **if equal it degenerates to a line segment**.
- $|z + 2| = |z - i| + \sqrt{5}$ (difference equals the distance between the two points): ray starting from i , with **closed endpoint at i** .
- $(z + i) \cdot \overline{\operatorname{conj}(z + i)} = 2 \Rightarrow$ centre $-i$, radius $\sqrt{2}$; $z - \bar{z} \leq 6i \Rightarrow \operatorname{Im} z \leq 3$.
- Mark-earning points when drawing: key points or centre, boundary position, solid or dashed lines, shaded region.

⑥ **Loci: write equation from graph**: use only z , not x, y ; circle $\Rightarrow |z - c| \leq r$; sector $\Rightarrow$ two $\arg$ inequalities + one modulus inequality; find angles exactly using trigonometry.

⑦ Maximum and minimum values on a locus

- Distance from a point on a circle to c : maximum $d + r$, minimum $|d - r|$ (d = distance from the centre to c). Nearest point on a ray: drop a perpendicular.
- The maximum argument occurs at the tangent: first use $\sin \theta = r/d$ to find the angle between the tangent and the line to the centre, then add the argument of the centre, paying attention to the range specified in the question.

⑧ Vectors and geometric methods

- $z + r$ (sum of two equal-length vectors) forms a rhombus $\Rightarrow \arg(z + r) = \theta/2$; $z - k$ forms a triangle, use the sine or cosine rule.
- Regular hexagon: first check where the centre is. When the centre is at the origin, the vertices have equal modulus and arguments differ by $\pi/3$; in 2025F-Q7 z_0 is at the origin, the moduli are $r, \sqrt{3}r, 2r, \sqrt{3}r, r$ respectively, and the product is $6r^5$.
- Trick: $\sum n \cdot i^n$ in groups of 4 terms, each group sums to $2 - 2i$; take modulus of $(a + bi)^3 \Rightarrow (a^2 + b^2)^3 = |\dots|^2$.

2 Functions and graphs

① **Domain of composite functions** ★ most frequently mentioned (6 years)

- $f(g(x))$ is defined $\Leftrightarrow x \in \operatorname{dom} g$ **and** $g(x) \in \operatorname{dom} f$. After substituting, don't simplify first; each restriction usually earns 1 mark.
- Sources of restrictions: denominator $\neq 0$; expression under a square root ≥ 0 ; **when the square root is in the denominator > 0** ; argument of $\ln > 0$.
- Example: $f = 1 - \sqrt{x - 4}$, $g = 1/x^2 \Rightarrow 1/x^2 \geq 4 \Rightarrow -\frac{1}{2} \leq x \leq \frac{1}{2}$ and $x \neq 0$. $f = \sqrt{4 - x}$, $g = 1/x^2 \Rightarrow x \leq -\frac{1}{2}$ or $x \geq \frac{1}{2}$.
- $\Delta \sqrt{x^2} = |x|$, so $f(h(g(x)))$ is not necessarily equal to $f(x)$.

② Inverse functions

- Exists $\Leftrightarrow$ one-to-one (horizontal line test); otherwise restrict the domain; for the "largest possible domain", go up to the vertex.
- Method: swap x and y (1 mark) $\rightarrow$ solve for $y \rightarrow$ the sign is determined by **ran $f^{-1} = \operatorname{dom} f$** ; $\operatorname{dom} f^{-1} = \operatorname{ran} f$.
- The graphs are symmetric about $y = x$; mark the corresponding endpoints $(a, b) \leftrightarrow (b, a)$. Whether $f^{-1}(-1)$ is defined: first check whether -1 is in $\operatorname{ran} f$.
- Inverse trigonometric functions: $x = 2 \tan y \Rightarrow y' = 2/(4 + x^2)$ (use implicit differentiation, then use $1 + \tan^2 y$ to change to x).
- Explaining "g is not one-to-one": "because $g(1) = g(-1) = 1$ ", don't write "it is many to one".

③ Graphing $y = 1/f(x)$ ★ often tested in CF

- Zero of $f \Rightarrow$ vertical asymptote; as $x \rightarrow \pm\infty$, $f \rightarrow \pm\infty \Rightarrow$ horizontal asymptote $y = 0$.
- Vertical asymptote of f at $x = b \Rightarrow 1/f$ has an open point at $(b, 0)$ (range excludes 0); horizontal asymptote of f at $y = c$ ($c \neq 0$) $\Rightarrow y = 1/c$.
- Local maximum $(a, m) \leftrightarrow$ local minimum $(a, 1/m)$; points where $f = \pm 1$ stay fixed; sign does not change; an open point of f at $(a, b) \Rightarrow$ an open point of $1/f$ at $(a, 1/b)$; if $b = 0$ it becomes a one-sided asymptote.
- Find the range: read from the graph of $1/f$, noting values that are not included.

④ Absolute value functions

- $|f(x)|$: reflect the part below the x-axis above the x-axis; $f(|x|)$: keep the right half and mirror it to the left; $f(-|x|)$: keep the left half and mirror it to the right.
- Number of solutions of $|f(x)| = k$: use the horizontal line $y = k$ to intersect; check endpoint k separately; you can draw the number-of-solutions function $N(k)$.
- The solution set of $|x + 1| + |x + a| = k$ is an interval $\Rightarrow$ the flat-bottom interval of the sum of two absolute values is the solution set ($a = 3, k = 2$).
- Determine $y = a|x - b| + c$ from the graph: the vertex gives b and c , the gradient gives a .

⑤ Rational functions: determine parameters from graph / sketch graph

- Zero $\leftrightarrow$ factor of numerator; vertical asymptote $\leftrightarrow$ factor of denominator; horizontal asymptote = ratio of leading coefficients; oblique asymptote = quotient from polynomial division ($x - 1 - 3/(x + 1) \Rightarrow y = x - 1$).
- Repeated factor: in the numerator $\Rightarrow$ graph touches the x-axis but does not cross; in the denominator $\Rightarrow$ both sides of the asymptote have the same sign (both tend to $-\infty$).
- Write the reason (1 mark): "c = 3 since $x = 3$ is a vertical asymptote". A quadratic with no zeros of the form $a(x - 3)^2 + c$ requires a and c to have the same sign.
- Sketching checklist (1 mark each): x-intercept, y-intercept, vertical asymptote, horizontal or oblique asymptote, turning point (mark if present), behaviour on either side of the asymptote. First check for cancelled factors (holes).

3 Three-dimensional vectors and systems of linear equations

① Systems of linear equations ★ Appeared in CF in 7 out of 10 years

- Three elimination steps (1 mark each): use two pairs of equations to eliminate the same variable $\rightarrow$ solve for one variable $\rightarrow$ back-substitute.
- With a parameter: reduce to $(k - 2)z = c - 5 \Rightarrow k \neq 2$: unique solution (c arbitrary); $k = 2$ and $c \neq 5$: no solution; $k = 2$ and $c = 5$: infinitely many solutions.
- Δ For infinitely many solutions, write the solution set (2022 report: many people stop after writing "infinitely many solutions"), expressing it using a parameter or relationships between variables. Example: $z = t \Rightarrow y = t - 1, x = (11 - 3t)/2$, i.e. $r = (11/2, -1, 0) + t(-3/2, 1, 1)$.
- Geometric interpretation: unique solution = three planes intersect at a point; one parameter = intersect in a line (two planes may coincide; both points must be written); two parameters = three planes coincide; no solution = there are parallel planes, or the three planes

intersect pairwise in three parallel lines (triangular prism, 2020F-Q4). First check whether the normal vectors are proportional.

② Equations of planes

- Parametric form $r = a + \lambda u + \mu v \Rightarrow n = u \times v$ (write each component of the cross product clearly) $\Rightarrow r \cdot n = a \cdot n$, i.e. $ax + by + cz = k$.
- Through three points: cross product of two edge vectors; $z = 2x + y + 4 \Rightarrow n = (2, 1, -1)$.
- Contains line L and is perpendicular to plane 1 $\Rightarrow n_2 = d_L \times n_1$; contains two intersecting lines $\Rightarrow n = d_1 \times d_2$.

③ Intersections, angles, distances

- Line and plane: substitute $r = a + \lambda d$, solve for λ , write down the point (3 marks). Line and sphere: obtain a quadratic equation in λ ; two solutions, tangent, no intersection; or compare the distance from the centre of the sphere to the line with the radius.
- Two lines: equate components, use two equations to solve for λ, μ , **substitute into the third to check**; if it does not hold, they do not intersect, and this step is the proof.
- Point on a plane closest to point L : construct a line through L along the normal vector and find its intersection with the plane.
- Line-plane angle: $\sin \theta = |d \cdot n| / (|d||n|)$. Distance from a point to a plane = $|n \cdot (P - A)| / |n|$, where A is any point on the plane; this formula is not on the formula sheet, so when used include a derivation or use the normal method. Volume of a parallelepiped = $|c \cdot (a \times b)|$.

④ Spheres: complete the square for the general form, e.g. $x^2 + y^2 + z^2 - 4x + 2y - 6z + 5 = 0 \Rightarrow |r - (2, -1, 3)| = 3$; endpoints of a diameter $\Rightarrow$ centre is the midpoint (report: often miscalculated); tangent to a plane $\Rightarrow$ radius = distance from the centre to the plane; passes through all vertices of a rectangular prism $\Rightarrow$ centre is the midpoint of the space diagonal.

⑤ Vector proofs (in both CF and CA): express the required vectors in terms of basis vectors $\rightarrow$ expand the dot product (key: get 4 terms) $\rightarrow$ use the given conditions (square $\Rightarrow a \cdot b = 0, |a| = |b|$) $\rightarrow$ conclude. Marks are awarded for notation.

⑥ Applications: reflection $\Rightarrow$ the sum of the unit vectors $B \rightarrow S$ and $B \rightarrow R \parallel n$; the steepest downhill direction on a slope lies in the slope plane and is perpendicular to the horizontal line; for distance problems involving moving objects, see the next section.

4 Vector calculus and motion (CA paper only)

① Differentiation: $v = r'$, $a = v'$, differentiate component by component; speed $|v|$; draw vectors on the graph (start at the position at that time). Constant speed $\neq$ constant velocity (direction is changing); you must write that $\sin^2 + \cos^2 = 1$ was used (1 mark).

② Integrating to find $r(t)$

- Integrate component by component, add a **constant vector**, and use $r(0)$ to determine each component (2018 report: many people do not write this clearly).
- Gravity + wind: $a = (0, -9.8) \Rightarrow r = ((70 \cos \theta + w)t, 70 \sin \theta t - 4.9t^2)$; range: set $y = 0$, solve for t , then substitute into x .
- Horizontal drag + vertical lift: $x' = 32e^{(-0.05t)}$, $h'' = s - 9.8$; use the given r to find s . For landing angle, use the angle between the velocity vector and the slope.

③ Eliminating the parameter to get the Cartesian equation ★

- $\sin^2 + \cos^2 = 1 \Rightarrow$ circle or ellipse; double-angle formulas directly give a relationship between x and $y \Rightarrow$ a segment of a parabola; first convert to the same angle then use $\sin^2 + \cos^2 \Rightarrow$ general curve (e.g. figure eight); $(e^t + e^{-t})^2 - (e^t - e^{-t})^2 = 4 \Rightarrow x^2 - y^2 = 4$; x is linear in $t \Rightarrow$ solve directly for t and substitute into y .
- Δ State the range of values, derived from the range of t ; often both x and y must be restricted: $x \geq 2$ and $y \geq 0$ (writing only $x \geq 2$ loses marks); $1 \leq x \leq e^3$.

④ Distance travelled and displacement

- Distance travelled $= \int_a^b \sqrt{\dot{x}^2 + \dot{y}^2} dt$; first determine the start and end times: the time for one lap = the least common multiple of the periods of the components ($2\pi, 4\pi, 6\pi$ have all been examined), or find t from the coordinates.
- Explanation: $\int_0^1 v dt =$ "the change in displacement (position vector) during the first second"; $\int |v| dt =$ distance travelled. The 2025 report specifically identified confusion between these two. Draw the displacement vector on the graph: from $r(a)$ to $r(b)$.

⑤ Closest distance and relative position

- Same time parameter: relative position $d(t) = r_P - r_Q$ (order matters when asked for relative position, 1 mark); closest $\Leftrightarrow d(t) \cdot (v_P - v_Q) = 0$; "how long within 50 m" $\Rightarrow$ solve $|d(t)| < 50$.
- Speed and departure time are both adjustable $\Rightarrow$ shortest distance between two lines: use independent parameters λ, μ , and make PQ perpendicular to both d_1 and d_2 .
- Angle of descent $= |90^\circ - \phi|$ (ϕ is the angle between the velocity and k), or use $\sin^{-1}(|v_z|/|v|)$ directly.

5 Integration techniques and applications

① Definite integrals with a given substitution ★ CF almost every year (4–7 marks)

- Five steps (1 mark each): find dx from the substitution $\rightarrow$ change the limits $\rightarrow$ convert the integrand completely into the new variable and simplify $\rightarrow$ integrate $\rightarrow$ substitute to get the exact value.
- Trigonometric substitution: for $\sqrt{a^2 - x^2}$ use $x = a \sin \theta$; for $x^2 + a^2$ use $x = a \tan \theta$. Example: $\int_0^{\pi/3} \sqrt{1 - x^2/4} dx$, $x = 2 \sin \theta \Rightarrow \int_0^{\pi/3} (\pi/3) 2 \cos^2 \theta d\theta = \pi/3 + \sqrt{3}/4$.
- Indefinite integral version ($\int x(1+x)^n dx$): do not change the limits; back-substitute into x at the end. Δ Simplifying after substitution is often incorrect (2020, 2023 reports).

② Trigonometric identities: $\cos^2 = \frac{1}{2}(1 + \cos 2x)$; $\sin^2 = \frac{1}{2}(1 - \cos 2x)$; $(\sin x + \cos x)^2 = 1 + \sin 2x$; product-to-sum $2 \sin A \cos B = \sin(A+B) + \sin(A-B)$; $\tan^2 = \sec^2 - 1$.

③ Partial fractions ★ CF high frequency

- Forms: distinct linear factors $A/(x-a) + B/(x-b)$; repeated root $A/(x+1) + B/(x+1)^2$; irreducible quadratic $(Bx+C)/(x^2+2)$. When the question specifies the form, write it as given.
- After putting over a common denominator, equate numerators (1 mark) $\rightarrow$ substitute special values or compare coefficients to find the constants (1 mark) $\rightarrow$ integrate term by term: $\ln|x-a|$, $-B/(x+1)$, when the numerator is the derivative of the denominator use $\ln|$ quadratic, $q/(x^2+4) \rightarrow (q/2)\tan^{-1}(x/2) \rightarrow$ write $+c$.

- Example: $1/(x^2-1) \Rightarrow \frac{1}{2} \ln|(x-1)/(x+1)| + c$; $(5x+3)/(x+1)^2 = 5/(x+1) - 2/(x+1)^2$. If no hint is given, factorise the denominator yourself first (1 mark).

④ Area

- First find intersection points and points of tangency; when writing the curve as $x = g(y)$ is more convenient, integrate with respect to y (key: marks are given for both methods).
- Symmetry: for an ellipse or a shape symmetric about the coordinate axes, multiply by a factor; circular segment: the difference between the upper and lower branches is $2\sqrt{\dots}$.
- A definite integral can be interpreted as a geometric area (triangle, trapezium). Comparing a numerical area with πr^2 can determine whether the curve is a circle.

⑤ Volumes of solids of revolution ★ CA high frequency

- About the x -axis $\pi \int y^2 dx$; about the y -axis $\pi \int x^2 dy$ (first express x^2 as a function of y); between two curves $\pi \int (R^2 - r^2)$, Δ not $\pi \int (R - r)^2$.
- The integral expression must include π , the limits, and dx or dy (the limits + notation together give 1 mark).
- Context: depth when filled to 80% $\Rightarrow$ set $\int_0^h \pi x^2 dy = 0.8V$ and solve for h ; filling rate $dV/dt = (dV/dh)(dh/dt)$; increment $\delta h \approx \delta V \div \pi x^2|_{(y=h)}$; annulus $R, r = 3 \pm \sqrt{4 - y^2}$.

6 Rates of Change and Differential Equations

① Implicit Differentiation

- Differentiate both sides with respect to x ; for terms containing y , multiply by y' ; use the product and chain rules as usual. If the question says not to solve for y' , don't; substitute the point directly to evaluate.
- Horizontal tangents: set numerator $= 0$, solve simultaneously with the original equation; second derivative: differentiate the expression already found again, substitute x, y, y' .

② Related Rates ★

- Write a relationship (eliminate extra variables if necessary; you can also keep two variables and differentiate implicitly) $\rightarrow$ differentiate with respect to $t \rightarrow$ substitute the values at that instant (often find θ, h first) $\rightarrow$ include units.
- Example: regular hexagon $A = (3\sqrt{3}/2)s^2$, $s = 4$, $ds/dt = 0.5 \Rightarrow dA/dt = 6\sqrt{3} \approx 10.39 \text{ cm}^2/\text{s}$. A lighthouse makes 3 revolutions per minute $\Rightarrow d\theta/dt = \pi/10 \text{ rad/s}$.
- Circular motion (Ferris wheel, merry-go-round): $d\theta/dt = 2\pi/T$; use cosine rule $s^2 = 89 - 80\cos \theta$ or $y = R \sin(\theta + \alpha)$; differentiate implicitly with respect to t ; to find the maximum ds/dt , differentiate again (geometrically, this is when PC is tangent to the circle). Δ Height above ground = height relative to centre $+ R$.

③ Slope Fields

- Substitute a point to find the slope (1 mark); infer the equation from the field: see whether the slope depends only on x or only on y , where it is 0 and where it is undefined.
- Sketch the solution curve: pass through the specified point, follow the short line segments, mark symmetry; where the denominator is 0 and the numerator $\neq 0$ the tangent is vertical, and the curve turns back there; where the slope is 0 there is a horizontal tangent; where the slope is greatest is a point of inflection (2025A-Q16 point of inflection at $(3, -1)$).

④ Separation of Variables

- Write as $\int g(y)dy = \int f(x)dx$ (1 mark) $\rightarrow$ integrate, add only one constant, put absolute value around $\ln \rightarrow$ substitute initial values $\rightarrow$ write in explicit form.
- $dP/dt = kP \Rightarrow P = P_0 e^{kt}$. Δ "instantaneous growth rate is 10%" means $dP/dt = 0.1P$ ($P_0 = 30000e^{-(4.5)} \approx 333$), not multiplying by 1.1 each year (≈ 412).
- Example: $dh/dt = -1/(100h) \Rightarrow h^2 = h_0^2 - t/50$; $dv/dt = -9.8 - 2v \Rightarrow v = 4.9(e^{-2t} - 1)$; for depth, integrate v again.

⑤ Logistic Growth ★

- $dP/dt = rP(k - P) \Leftrightarrow P = k/(1 + Ae^{(-rkt)})$: carrying capacity = the non-zero P that makes $dP/dt = 0$; **growth is fastest when $P = k/2$** , maximum growth rate $= r(k/2)^2$.
- Δ Letters depend on the question (in some questions k is the proportionality constant; in others it is written as $(1/r)P(k - P)$). Example: $P = 18000/(10.25e^{(-0.15t)} + 1)$ $\Rightarrow$ carrying capacity 18000, maximum growth rate $0.15 \times 9000 \times \frac{1}{2} = 675$ horses/year.
- Sketching: $k/2 < P_0 < k \Rightarrow$ concave down throughout, no point of inflection; $P_0 > k \Rightarrow$ decreasing towards k ; changing the carrying capacity $\Rightarrow$ new point of inflection and asymptote.
- Explanation: when $P < k/2$ the growth rate increases as P increases; when $P > k/2$ it decreases and approaches k . For equations that are not logistic (e.g. $0.001(100 - P)^2$), analyse the change in slope directly.

⑥ Velocity as a Function of Displacement $v = f(x)$ ★★

named three times in the report

- $a = dv/dt = v \cdot dv/dx = d(\frac{1}{2}v^2)/dx$. Δ dv/dx itself is not acceleration. Example: $v = -0.2x \Rightarrow a = 0.04x$; $v = x/8 \Rightarrow a = x/64$.
- $x(t)$: $dx/dt = v(x)$, separate variables, substitute the initial position. Example: $v = x/8$, $x(0) = 192 \Rightarrow x = 192e^{(t/8)}$.
- To decide SHM: only $a = -n^2(x - c)$ (proportional to displacement and opposite in direction) qualifies; $a = 0.04x$ is not.

⑦ Simple Harmonic Motion SHM

- $\ddot{x} = -n^2x \Rightarrow x = A \cos(nt + \phi)$; $T = 2\pi/n$; $v_{\max} = An$; $v^2 = n^2(A^2 - x^2)$.
- Determine A and phase: $x(0) = 0$ and $v(0) > 0 \Rightarrow A \sin(nt)$; if no direction is given, both $\pm A \sin$ receive marks; $n = 2$, $x(0) = -1$, $v(0)^2 = 32 \Rightarrow A = 3$.
- Distance travelled: $4A$ per period, $2A$ per half-period; if less than half a period and the starting point is not at the equilibrium position or an endpoint, use $\int |v| dt$.
Damping: $A = A_0 e^{(-kt)}$; $A < 0.01$ is treated as stopped.

- **Increment Formula:** $\delta y \approx (dy/dx)\delta x$; $\delta N \approx (dN/dt) \cdot \delta t$ (dN/dt is given directly by the equation; watch the time units).

region if required.

② Finding Sample Size

- First sketch to decide whether it is two-tailed, one-tailed or half-interval, then find the critical z : $P(|X - \mu| < d) = 0.98 \Rightarrow z = 2.326$; $P(X > 25) = 0.03 \Rightarrow z = 1.881$; $P(0 < Z < k) = 0.4 \Rightarrow k = 1.282$.
- $n = (z\sigma/d)^2$: when "at least" is required, round up ($304.3 \Rightarrow 305$, $600.25 \Rightarrow 601$); when working backwards from an interval to n , take the nearest integer (differences caused by using $z = 2.576$ and 2.58 are both accepted by the key).

③ Confidence Intervals

- $\bar{x} \pm z \cdot s/\sqrt{n}$. Work backwards: midpoint $= \bar{x}$; half-width $= z \cdot \sigma_X$ ($150 \leq \mu \leq 200 \Rightarrow \sigma_X = 25/1.96 = 12.76$); find s from the width and n .
- Changing sample size: width $\propto 1/\sqrt{n}$ ($n \times 16 \Rightarrow$ width becomes $1/4$); $2n \Rightarrow$ standard error divided by $\sqrt{2}$.

④ Interpreting Confidence Intervals ★ must memorise (state True / False first, then give a reason)

- μ is fixed but unknown; a particular interval either contains μ or it does not, **and you cannot know which**.
- 95% means that under repeated sampling about 95% of intervals contain μ , **not** "the probability that μ lies in this interval is 95%" .
- The next sample mean will not necessarily fall in the interval; a single individual even less so (the variation in individuals is far greater than the variation in sample means).
- A wider interval does not mean it is more likely to contain μ ; 10 out of 50 intervals being on the low side is normal variation; but a large number of intervals not containing μ (far more than 5%, e.g. 298 out of 500) is evidence, although you still cannot determine the cause.
- Copying the same sample once to make up $2n$ is not a new random sample, and the interval does not truly become narrower; combining two independent random samples is acceptable and gives higher precision.
- Intervals from different sample sizes cannot be compared directly in this way; a lower sample mean alone also cannot show that the new method is better.

- **Using an Interval to Test a Claim:** construct a μ interval for the new sample (e.g. $182.5 \pm 1.96 \times 15/6 = (177.6, 187.4)$) $\rightarrow$ whether the claimed value 175 lies in the interval $\rightarrow$ draw a conclusion. You can only draw a conclusion about whether the means are different; you cannot infer the source of the sample.

- **Factors Affecting Precision:** increasing $n \Rightarrow \sigma_X$ decreases, the interval becomes narrower, and the tail probability on the side away from μ decreases; increasing the confidence level $\Rightarrow$ the interval becomes wider; increasing σ (or s) $\Rightarrow$ the interval becomes wider.

7 Statistical Inference (only in the CA paper)

① Distribution and Probability of X ★

- "By the CLT, since n is large, X is approximately normal with mean μ and SD $\sigma/\sqrt{n}$ " (normal, mean, standard deviation, 3 marks in total). If the population is normal, this holds for any n .
- Population not normal (uniform, exponential, skewed): same method; if asked "does the answer change?" , answer no + CLT reason.
- Sum question: total of 50 observations < 8.96 kL $\Leftrightarrow X < 179.2$ L. Sketch the distribution of X : a narrow bell shape centred at μ , width about $\pm 3\sigma/\sqrt{n}$; mark the probability